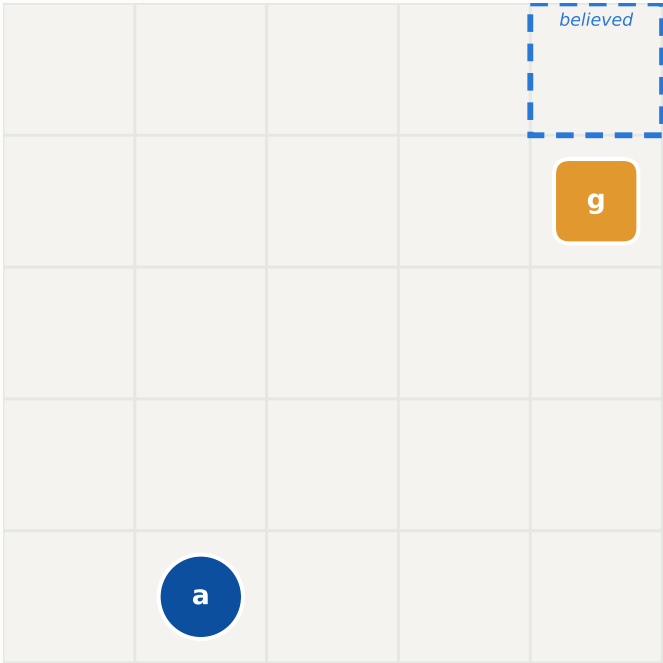


The discovered belief compound passes the false-belief test

On a held-out Sally-Anne scene, the learned fork(derive, commit) abstraction sends the agent to where it believes the goal is; the shortest program in the non-mental fragment sends it to the true goal — the answer three-year-olds give. (24/24 held-out scenes, both.)

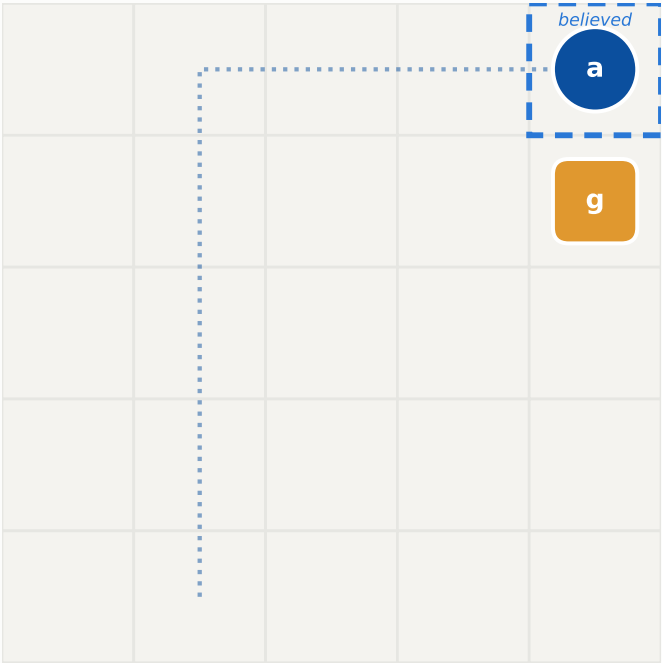
A held-out scene

agent a, true goal g; the goal is "believed" one cell away



Belief compound ✓

learner with the discovered fork(derive, commit)

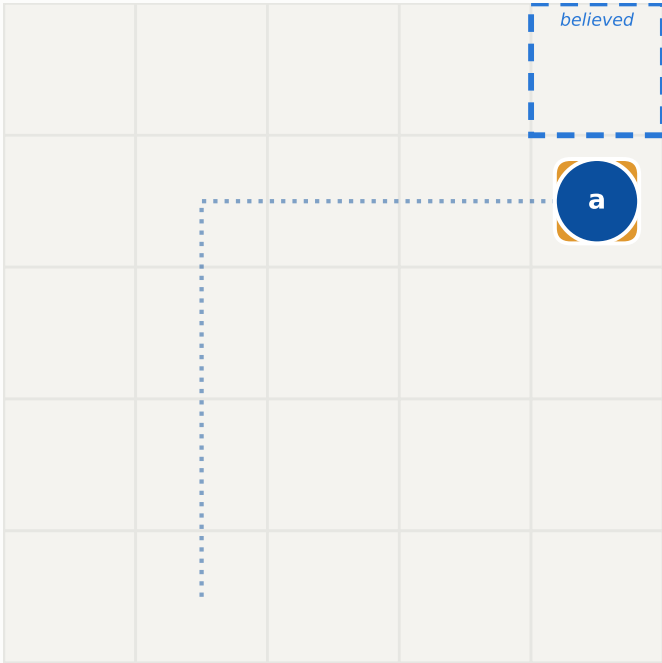


searches the BELIEVED cell — passes

```
(fork (compose (step 2 up) (optimize (neg_dist 2) 1)) (sync_to_world 1))
22.7 nats
```

Non-mental fragment ✗

learner restricted to physics only



searches the TRUE cell — the naive answer

```
(optimize (neg_dist 2) 1)
7.6 nats · shortest non-mental
```